

EXPERIMENT – 08

Q1. Swap Two Numbers (With and Without a Third Variable)

Problem Statement: Write a menu-driven C program to input two numbers and swap them using a third variable and also without using a third variable. Display the values before and after swapping.

Algorithm

1. Start.
2. Declare integer variables a, b, temp and choice.
3. Display the menu: 1. Swap using third variable, 2. Swap without using third variable, 3. Exit.
4. Read the value of choice.
5. If choice = 1, read a and b, display them, swap them using a third variable temp, and display a and b again.
6. If choice = 2, read a and b, display them, swap them without using a third variable ($a = a+b$, $b = a-b$, $a = a-b$), and display a and b again.
7. If choice = 3, display an exit message and stop the loop.
8. If choice is none of the above, display an invalid choice message.
9. Repeat steps 3 to 8 until choice = 3.
10. Stop.

Pseudocode

```
START
DECLARE a, b, temp, choice

REPEAT
    DISPLAY menu (1: swap using 3rd variable, 2: swap without 3rd variable, 3: exit)
    READ choice

    IF choice = 1 THEN
        READ a, b
        DISPLAY "Before swapping", a, b
        temp <- a
        a <- b
        b <- temp
        DISPLAY "After swapping", a, b

    ELSE IF choice = 2 THEN
        READ a, b
        DISPLAY "Before swapping", a, b
        a <- a + b
        b <- a - b
        a <- a - b
        DISPLAY "After swapping", a, b

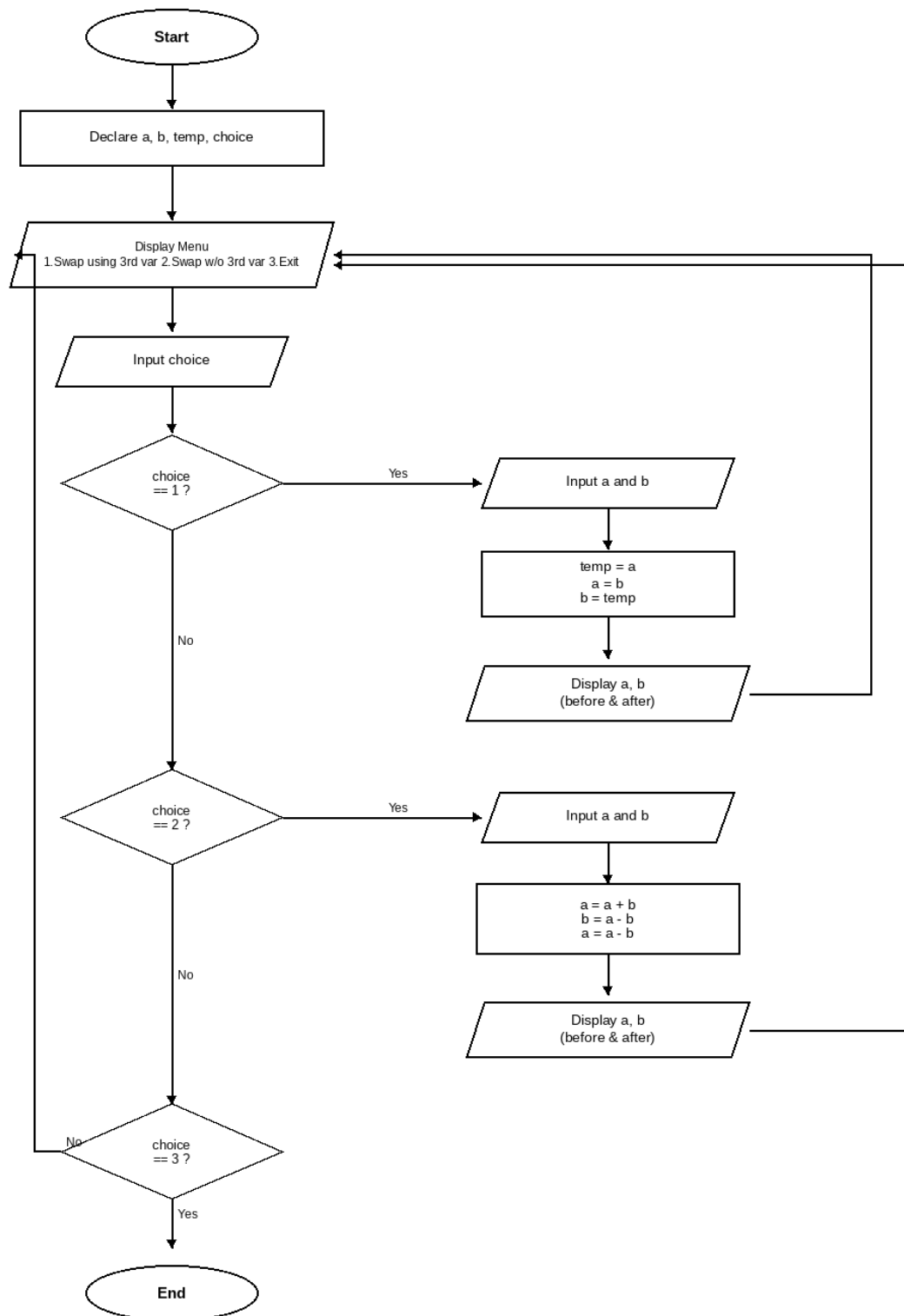
    ELSE IF choice = 3 THEN
        DISPLAY "Exiting program"

    ELSE
        DISPLAY "Invalid choice"

    END IF

UNTIL choice = 3
STOP
```

Flowchart



Test Case Table

Test No.	Input	Expected Output	Actual Output	Result
1	choice=1, a=5, b=10	Before: a=5,b=10 After: a=10,b=5	Before: a=5,b=10 After: a=10,b=5	Pass
2	choice=2, a=-3, b=8	Before: a=-3,b=8 After: a=8,b=-3	Before: a=-3,b=8 After: a=8,b=-3	Pass
3	choice=1, a=0, b=0	Before: a=0,b=0 After: a=0,b=0	Before: a=0,b=0 After: a=0,b=0	Pass
4	choice=3	Exiting program.	Exiting program.	Pass

C Program

```
#include<stdio.h>
```

```
int main(void)
```

```
{
```

```
    int a,b,temp,choice;
```

```
    do
```

```
    {
```

```
        printf("\n----- MENU ----- \n");
```

```
        printf("1. Swap using third variable\n");
```

```
        printf("2. Swap without using third variable\n");
```

```
        printf("3. Exit\n");
```

```
        printf("Enter your choice: ");
```

```
        scanf("%d",&choice);
```

```
        switch(choice)
```

```
        {
```

```
            case 1:
```

```
                printf("Enter value of a: ");
```

```
                scanf("%d",&a);
```

```
                printf("Enter value of b: ");
```

```
                scanf("%d",&b);
```

```
                printf("\nBefore swapping: a = %d, b = %d\n",a,b);
```

```
                temp=a;
```

```
                a=b;
```

```
                b=temp;
```

```
                printf("After swapping: a = %d, b = %d\n",a,b);
```

```
                break;
```

```
            case 2:
```

```
                printf("Enter value of a: ");
```

```
                scanf("%d",&a);
```

```
                printf("Enter value of b: ");
```

```
                scanf("%d",&b);
```

```
                printf("\nBefore swapping: a = %d, b = %d\n",a,b);
```

```
                a=a+b;
```

```
                b=a-b;
```

```
                a=a-b;
```

```
                printf("After swapping: a = %d, b = %d\n",a,b);
```

```
                break;
```

```
            case 3:
```

```
                printf("Exiting program.\n");
```

```
                break;
```

```
            default:
```

```

        printf("Invalid choice, try again.\n");
    }
}while(choice!=3);

return 0;
}

```

Compilation, Execution and Output

```

adilshakil - zsh - 120x48

adilshakil@ADILs-MacBook-Air demo % gcc day9_swap.c -o day9_swap
adilshakil@ADILs-MacBook-Air demo % ./day9_swap

----- MENU -----
1. Swap using third variable
2. Swap without using third variable
3. Exit
Enter your choice: 1
Enter value of a: 5
Enter value of b: 10

Before swapping: a = 5, b = 10
After swapping: a = 10, b = 5

----- MENU -----
1. Swap using third variable
2. Swap without using third variable
3. Exit
Enter your choice: 2
Enter value of a: -3
Enter value of b: 8

Before swapping: a = -3, b = 8
After swapping: a = 8, b = -3

----- MENU -----
1. Swap using third variable
2. Swap without using third variable
3. Exit
Enter your choice: 1
Enter value of a: 0
Enter value of b: 0

Before swapping: a = 0, b = 0
After swapping: a = 0, b = 0

----- MENU -----
1. Swap using third variable
2. Swap without using third variable
3. Exit
Enter your choice: 3
Exiting program.

adilshakil@ADILs-MacBook-Air demo %

```

Q2. Sum of First n Natural Numbers and Sum of Squares

Problem Statement: Write a menu-driven C program to find the sum of the first n natural numbers and the sum of squares of the first n natural numbers.

Algorithm

1. Start.
2. Declare integer variables n, i, choice and sum.
3. Display the menu: 1. Sum of first n natural numbers, 2. Sum of squares of first n natural numbers, 3. Exit.
4. Read the value of choice.
5. If choice = 1, read n, initialise sum = 0, add i to sum for i = 1 to n, and display sum.
6. If choice = 2, read n, initialise sum = 0, add $i*i$ to sum for i = 1 to n, and display sum.
7. If choice = 3, display an exit message and stop the loop.
8. If choice is none of the above, display an invalid choice message.
9. Repeat steps 3 to 8 until choice = 3.
10. Stop.

Pseudocode

START

DECLARE n, i, choice, sum

REPEAT

 DISPLAY menu (1: sum of first n natural numbers, 2: sum of squares, 3: exit)

 READ choice

 IF choice = 1 THEN

 READ n

 sum <- 0

 FOR i <- 1 TO n

 sum <- sum + i

 END FOR

 DISPLAY sum

 ELSE IF choice = 2 THEN

 READ n

 sum <- 0

 FOR i <- 1 TO n

 sum <- sum + (i * i)

 END FOR

 DISPLAY sum

 ELSE IF choice = 3 THEN

 DISPLAY "Exiting program"

 ELSE

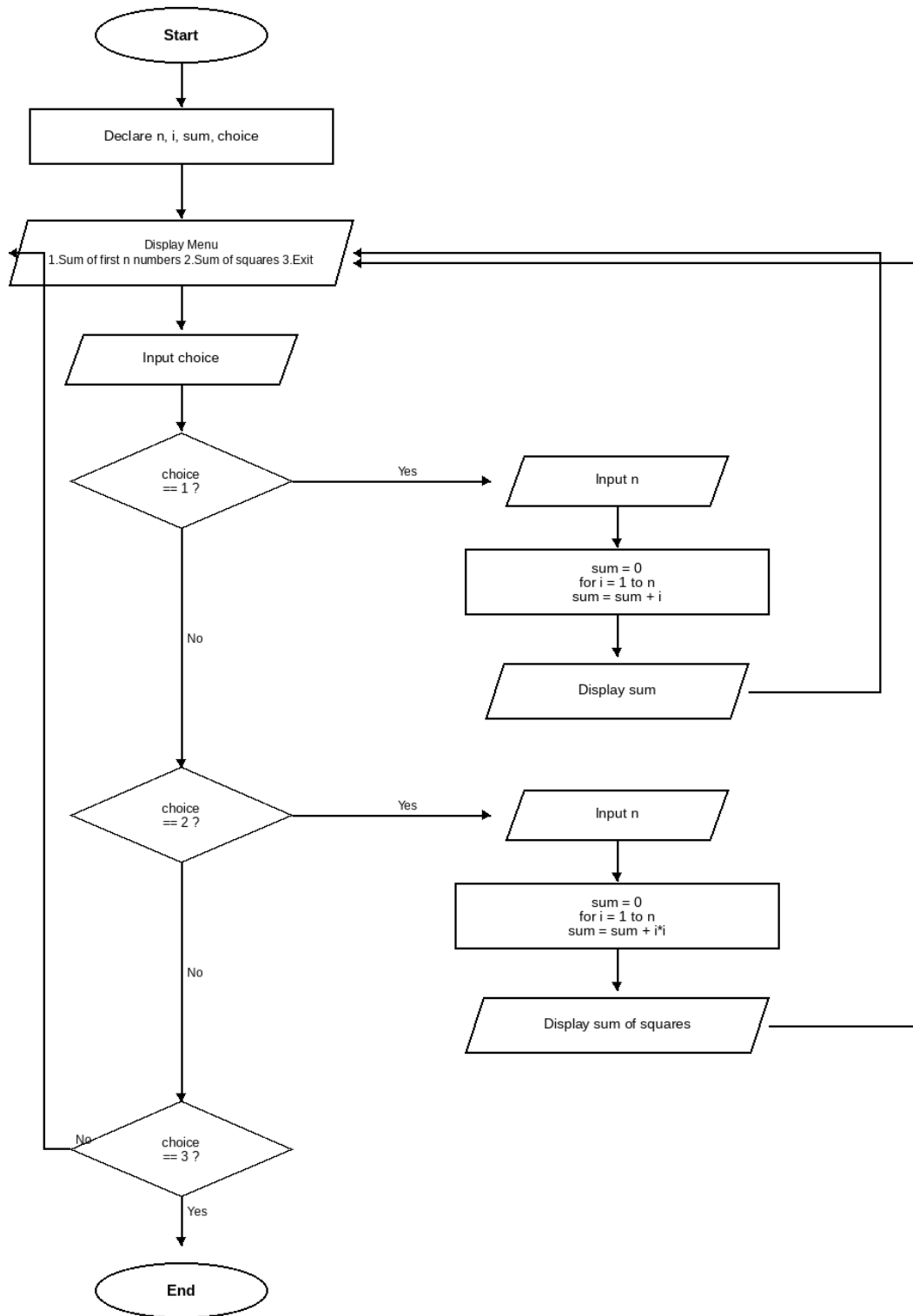
 DISPLAY "Invalid choice"

 END IF

UNTIL choice = 3

STOP

Flowchart



Test Case Table

Test No.	Input	Expected Output	Actual Output	Result
1	choice=1, n=5	Sum = 15	Sum = 15	Pass
2	choice=2, n=5	Sum of squares = 55	Sum of squares = 55	Pass
3	choice=1, n=1	Sum = 1	Sum = 1	Pass
4	choice=2, n=1	Sum of squares = 1	Sum of squares = 1	Pass
5	choice=1, n=0	Sum = 0	Sum = 0	Pass
6	choice=3	Exiting program.	Exiting program.	Pass

C Program

```
#include<stdio.h>
```

```
int main(void)
{
    int n,i,choice;
    long sum;

    do
    {
        printf("\n----- MENU ----- \n");
        printf("1. Sum of first n natural numbers\n");
        printf("2. Sum of squares of first n natural numbers\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d",&choice);

        switch(choice)
        {
            case 1:
                printf("Enter value of n: ");
                scanf("%d",&n);

                sum=0;
                for(i=1;i<=n;i++)
                    sum=sum+i;

                printf("Sum of first %d natural numbers = %ld\n",n,sum);
                break;

            case 2:
                printf("Enter value of n: ");
                scanf("%d",&n);

                sum=0;
                for(i=1;i<=n;i++)
                    sum=sum+(i*i);

                printf("Sum of squares of first %d natural numbers = %ld\n",n,sum);
                break;

            case 3:
                printf("Exiting program.\n");
                break;

            default:
                printf("Invalid choice, try again.\n");
        }
    }while(choice!=3);

    return 0;
```

}

Compilation, Execution and Output

```
adilshakil - zsh - 120x55

adilshakil@ADILs-MacBook-Air demo % gcc day9_sum.c -o day9_sum
adilshakil@ADILs-MacBook-Air demo % ./day9_sum

----- MENU -----
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 1
Enter value of n: 5
Sum of first 5 natural numbers = 15

----- MENU -----
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 2
Enter value of n: 5
Sum of squares of first 5 natural numbers = 55

----- MENU -----
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 1
Enter value of n: 1
Sum of first 1 natural numbers = 1

----- MENU -----
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 2
Enter value of n: 1
Sum of squares of first 1 natural numbers = 1

----- MENU -----
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 1
Enter value of n: 0
Sum of first 0 natural numbers = 0

----- MENU -----
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 3
Exiting program.

adilshakil@ADILs-MacBook-Air demo %
```


Q3. Simple Interest, Compound Amount, Compound Interest and Total Payable Amount

Problem Statement: Write a menu-driven C program to calculate the simple interest, compound amount, compound interest and total payable amount for a given principal, rate and time.

Algorithm

1. Start.
2. Declare float variables p, r, t, si, ca, ci, total and integer choice.
3. Display the menu: 1. Simple Interest, 2. Compound Amount, 3. Compound Interest, 4. Total Payable Amount, 5. Exit.
4. Read the value of choice.
5. If choice lies between 1 and 4, read the principal p, rate r and time t.
6. If choice = 1, compute $si = (p * r * t) / 100$ and display si.
7. If choice = 2, compute $ca = p * (1 + r/100)^t$ and display ca.
8. If choice = 3, compute $ca = p * (1 + r/100)^t$, $ci = ca - p$ and display ci.
9. If choice = 4, compute $si = (p * r * t) / 100$, $total = p + si$ and display total.
10. If choice = 5, display an exit message and stop the loop.
11. If choice is none of the above, display an invalid choice message.
12. Repeat steps 3 to 11 until choice = 5.
13. Stop.

Pseudocode

START

DECLARE p, r, t, si, ca, ci, total, choice

REPEAT

 DISPLAY menu (1: Simple Interest, 2: Compound Amount,
 3: Compound Interest, 4: Total Payable Amount, 5: Exit)

 READ choice

 IF choice >= 1 AND choice <= 4 THEN

 READ p, r, t

 END IF

 IF choice = 1 THEN

$si \leftarrow (p * r * t) / 100$

 DISPLAY si

 ELSE IF choice = 2 THEN

$ca \leftarrow p * (1 + r/100)^t$

 DISPLAY ca

 ELSE IF choice = 3 THEN

$ca \leftarrow p * (1 + r/100)^t$

$ci \leftarrow ca - p$

 DISPLAY ci

 ELSE IF choice = 4 THEN

$si \leftarrow (p * r * t) / 100$

$total \leftarrow p + si$

 DISPLAY total

 ELSE IF choice = 5 THEN

```
        DISPLAY "Exiting program"
```

```
    ELSE
```

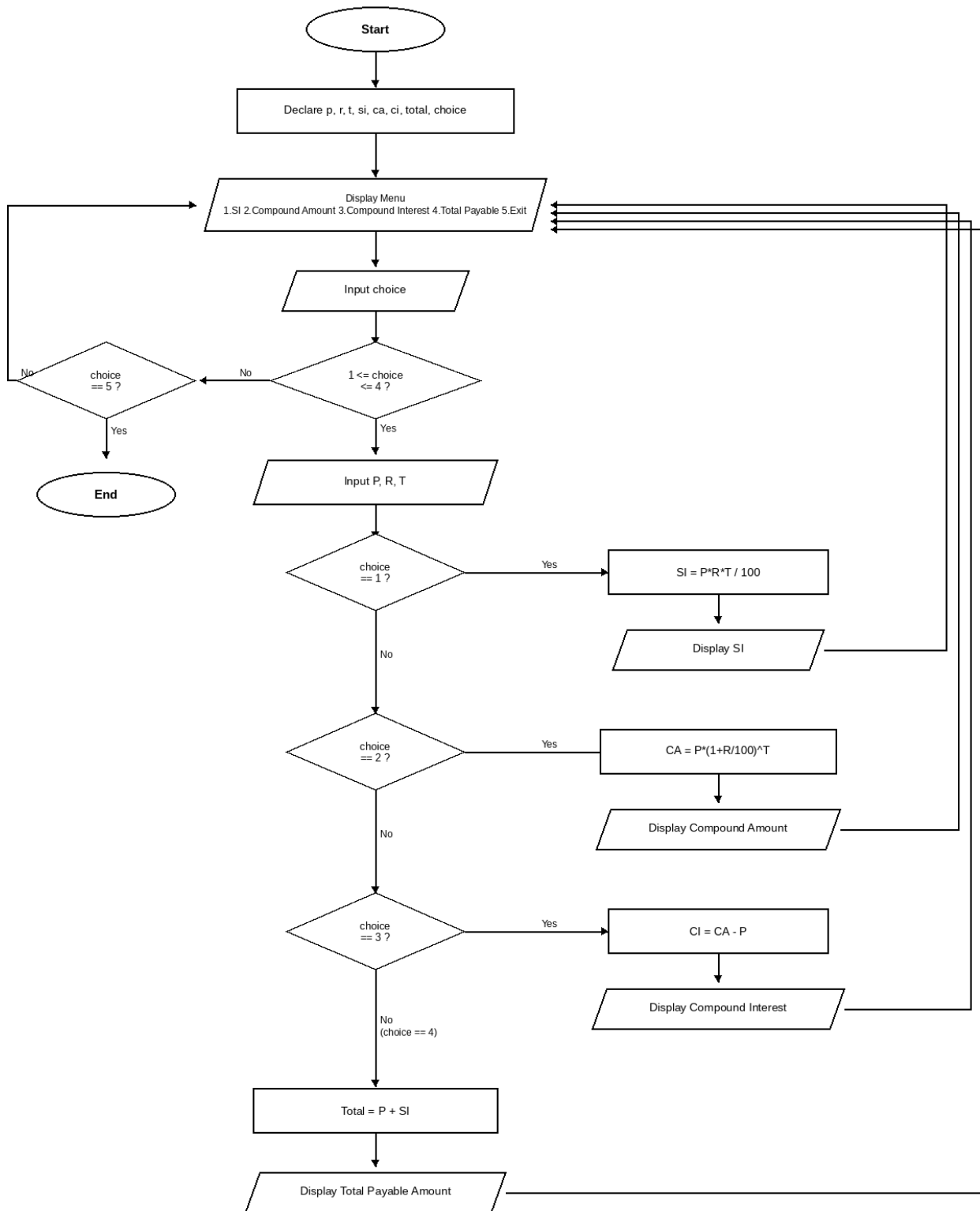
```
        DISPLAY "Invalid choice"
```

```
    END IF
```

```
UNTIL choice = 5
```

```
STOP
```

Flowchart



Test Case Table

Test No.	Input	Expected Output	Actual Output	Result
1	choice=1, P=1000, R=5, T=2	Simple Interest = 100.00	Simple Interest = 100.00	Pass
2	choice=2, P=1000, R=5, T=2	Compound Amount = 1102.50	Compound Amount = 1102.50	Pass
3	choice=3, P=1000, R=5, T=2	Compound Interest = 102.50	Compound Interest = 102.50	Pass
4	choice=4, P=1000, R=5, T=2	Total Payable Amount = 1100.00	Total Payable Amount = 1100.00	Pass
5	choice=5	Exiting program.	Exiting program.	Pass

C Program

```
#include<stdio.h>
#include<math.h>

int main(void)
{
    float p,r,t,si,ca,ci,total;
    int choice;

    do
    {
        printf("\n----- MENU ----- \n");
        printf("1. Simple Interest\n");
        printf("2. Compound Amount\n");
        printf("3. Compound Interest\n");
        printf("4. Total Payable Amount\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d",&choice);

        if(choice>=1 && choice<=4)
        {
            printf("Enter principal amount: ");
            scanf("%f",&p);
            printf("Enter rate of interest: ");
            scanf("%f",&r);
            printf("Enter time (in years): ");
            scanf("%f",&t);
        }

        switch(choice)
        {
            case 1:
                si=(p*r*t)/100;
                printf("Simple Interest = %.2f\n",si);
                break;

            case 2:
                ca=p*pow((1+r/100),t);
                printf("Compound Amount = %.2f\n",ca);
                break;

            case 3:
                ca=p*pow((1+r/100),t);
                ci=ca-p;
                printf("Compound Interest = %.2f\n",ci);
                break;

            case 4:
```

```

        si=(p*r*t)/100;
        total=p+si;
        printf("Total Payable Amount = %.2f\n",total);
        break;

    case 5:
        printf("Exiting program.\n");
        break;

    default:
        printf("Invalid choice, try again.\n");
    }
}while(choice!=5);

return 0;
}

```

Compilation, Execution and Output

```
adilshakil@ADILs-MacBook-Air demo % gcc day9_interest.c -o day9_interest -lm
```

```
adilshakil@ADILs-MacBook-Air demo % ./day9_interest
```

```
----- MENU -----
```

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit

Enter your choice: 1

Enter principal amount: 1000

Enter rate of interest: 5

Enter time (in years): 2

Simple Interest = 100.00

```
----- MENU -----
```

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit

Enter your choice: 2

Enter principal amount: 1000

Enter rate of interest: 5

Enter time (in years): 2

Compound Amount = 1102.50

```
----- MENU -----
```

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit

Enter your choice: 3

Enter principal amount: 1000

Enter rate of interest: 5

Enter time (in years): 2

Compound Interest = 102.50

```
----- MENU -----
```

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit

Enter your choice: 4

Enter principal amount: 1000

Enter rate of interest: 5

Enter time (in years): 2

Total Payable Amount = 1100.00

```
----- MENU -----
```

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit

Enter your choice: 5

Exiting program.

```
adilshakil@ADILs-MacBook-Air demo %
```